Stuti Varshney

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PROFESSIONAL SUMMARY

Full Stack Developer and B.Tech Information Technology graduate (2026) with hands-on experience building and deploying MERN applications, including EstateX, a live real estate platform with JWT authentication, REST APIs, and MongoDB Atlas. Completed internships in full-stack development and machine learning, where I built and deployed an XGBoost-based intrusion detection system (99.89% accuracy) as a Flask web application. National Semi-Finalist at Flipkart GRID 7.0. Seeking a Software Developer or MERN Stack Developer role.

TECHNICAL SKILLS

Languages: JavaScript, Python, C++, SQL
Frontend: React.js, HTML, CSS, Tailwind CSS, Bootstrap, jQuery, React Router
Backend: Node.js, Express.js, REST APIs, JWT Authentication, Mongoose
Databases: MongoDB Atlas, PostgreSQL
ML & Data: Scikit-learn, XGBoost, Pandas, NumPy, PyTorch, OpenCV, Flask
Tools: Git, GitHub, Postman, VS Code, Netlify, Render

PROJECTS

EstateX – Full-Stack Real Estate Platform May 2025 – Jul 2025
React.js, Vite, Tailwind CSS, Node.js, Express.js, MongoDB Atlas, Mongoose, JWT | Live Demo

- Built a full-stack real estate platform where users browse properties, view detailed listings, save favorites, send agent inquiries, and schedule visits (date and time slot) from a personalized dashboard.
- Developed a RESTful backend with Node.js and Express.js across 6 route modules (auth, properties, favorites, inquiries, visits, dashboard), using Mongoose schemas with ObjectId references for users, properties, inquiries, and visits on MongoDB Atlas.
- Implemented secure authentication with bcrypt password hashing, JWT-based login, and protected routes via custom middleware, with persistent sessions on the frontend using React Context API.
- Built a responsive UI with reusable React components and React Router for dynamic property pages; deployed the frontend on Netlify and the backend on Render, configuring environment variables, CORS, and Atlas network access.

SafeStride – Road Surface Defect Detection using YOLOv8 Jan 2025 – Apr 2025
Independent Research Project | Python, YOLOv8, PyTorch, OpenCV, Roboflow

- Improved YOLOv8 for detecting road defects (cracks and potholes) by adding SPD layers, non-strided convolutions and Faster-Net blocks for better small-defect detection.
- Trained and fine-tuned YOLOv8s on 5,986 road images from the India subset of RDD2022, annotating bounding boxes for 2 classes (cracks, potholes) in Roboflow, with a 7:2:1 train/val/test split.
- Optimized bounding box regression with a Wise-IoU loss, reaching 77.4% mAP at 43 FPS, a 2.8% improvement over baseline YOLOv8.

EXPERIENCE

Data Analytics Intern, CETPA Infotech Pvt. Ltd. – Noida, India Jan 2026 – Jun 2026

- Built a Machine Learning-based Network Intrusion Detection System (NIDS) on the CICIDS2017 dataset (2.83M records, 78 features), classifying 12 attack types including DDoS, PortScan, Bot, and brute-force attacks.
- Cleaned and preprocessed data, removing 307K+ duplicate records and handling missing/infinite values, reducing to 2.52M high-quality records for training.
- Trained and benchmarked Random Forest and XGBoost classifiers; XGBoost achieved 99.89% accuracy and was selected for deployment.
- Deployed the model via a Flask web application with an interactive Chart.js dashboard supporting CSV upload, real-time classification, threat-severity scoring, and downloadable PDF/CSV reports.
- Validated the system across 21 test cases (unit, functional, integration, UI, security) with a 100% pass rate.

Full Stack Development Intern, Bharat Intern – Remote, India Jul 2024 – Aug 2024

- Built StreamFlix, a responsive movie/TV discovery web app integrating the TMDB REST API to dynamically render trending, popular, top-rated, and upcoming titles, with client-side search and a movie-details modal.
- Implemented a "My List" feature using LocalStorage for persistent, refresh-safe state, plus loading skeletons and error handling for failed API calls.
- Deployed StreamFlix via GitHub Pages with Git-based version control; isolated API configuration from source using .gitignore.

EDUCATION

Banasthali Vidyapith – B.Tech. in Information Technology Jul 2022 – Jun 2026
CGPA: 7.73/10.0

Coursework: Data Structures and Algorithms, DBMS, Operating Systems, Computer Networks, Machine Learning, Object-Oriented Programming

ACHIEVEMENTS & PROGRAMS

- National Semi-Finalist, Flipkart GRID 7.0 (2025);
- Participant, Flipkart Girls Wanna Code 6.0 (2025).
- Completed the Include<Her> Cohort 3.0 DevOps program under SheBuildsOnAzure (2023).